

Full-Spectrum Spatial Pipeline

● PHASE 1: INPUT

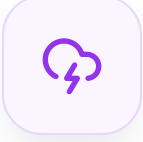


Edge Acquisition

Unskilled factory worker records a quick 4K orbital video of the physical part on any standard smartphone.

- ✓ **60FPS EXTRACTION**
Extracts high-fidelity frames instantly.
- ✓ **ZERO HARDWARE**
No expensive laser scanners needed.

● PHASE 2: TRANSFER



Encrypted Uplink

Data is securely transmitted from the MSME factory floor to the high-performance AWS/GCP GPU cluster.

- ✓ **TLS 1.3 SECURITY**
Bank-grade encryption for proprietary designs.
- ✓ **LOW-LATENCY**
Optimized packet routing to edge servers.

● PHASE 3: CORE AI



NeRF Spatial Engine

Frontier neural networks extrapolate perfect 3D depth and geometry strictly from the flat 2D video frames.

- ✓ **GAUSSIAN SPLATTING**
Ultra-fast volumetric scene rendering.
- ✓ **POINT CLOUD GEN**
Calculates millions of precise spatial coordinates.

● PHASE 4: COMPUTE



Mesh Synthesis

The raw neural point cloud is converted into a mathematically perfect, watertight polygonal mesh.

- ✓ **TOPOLOGY SEALING**
AI automatically closes optical gaps/holes.
- ✓ **UNIVERSAL EXPORT**
Outputs industry-standard .STEP / .STL formats.

● PHASE 5: OUTPUT



CAM Execution

The synthesized digital twin is sent directly to the factory floor's existing machining centers.

- ✓ **CNC INTEGRATION**
Directly fed into lathes, mills, or 3D printers.
- ✓ **INSTANT ROI**
Days of manual CAD drafting eliminated.